

Original LCTRSs

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times bool \times bool \times int \Rightarrow State \\ c00 : Loc \\ c10 : Loc \\ c20 : Loc \\ crit0 : Loc \\ c01 : Loc \\ c11 : Loc \\ c21 : Loc \\ crit1 : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll} 1 : & \text{state}(c00, p1, b0, b1, k) \rightarrow \text{state}(c10, p1, b01, b1, k) \quad [\neg(b01)] \\ 2 : & \text{state}(c10, p1, b0, b1, k) \rightarrow \text{state}(c20, p1, b0, b1, k) \quad [k \neq 0] \\ 3 : & \text{state}(c10, p1, b0, b1, k) \rightarrow \text{state}(crit0, p1, b0, b1, k) \quad [k = 0] \\ 4 : & \text{state}(c20, p1, b0, b1, k) \rightarrow \text{state}(c10, p1, b0, b1, k1) \quad [b1 \wedge k1 = 0] \\ 5 : & \text{state}(crit0, p1, b0, b1, k) \rightarrow \text{state}(c00, p1, b01, b1, k) \quad [b01] \\ 6 : & \text{state}(p0, c01, b0, b1, k) \rightarrow \text{state}(p0, c11, b0, b11, k) \quad [\neg(b11)] \\ 7 : & \text{state}(p0, c11, b0, b1, k) \rightarrow \text{state}(p0, c21, b0, b1, k) \quad [k \neq 1] \\ 8 : & \text{state}(p0, c11, b0, b1, k) \rightarrow \text{state}(p0, crit1, b0, b1, k) \quad [k = 1] \\ 9 : & \text{state}(p0, c21, b0, b1, k) \rightarrow \text{state}(p0, c11, b0, b1, k1) \quad [b0 \wedge k1 = 1] \\ 10 : & \text{state}(p0, crit1, b0, b1, k) \rightarrow \text{state}(p0, c01, b0, b11, k) \quad [b11] \\ 11 : & \text{state}(p0, p1, b0, b1, k) \rightarrow \text{success} \\ 12 : & \text{state}(crit0, crit1, b0, b1, k) \rightarrow \text{error} \end{array} \right\}$$

APR Problem

$$\{ \quad 13 : \quad \text{state}(c00, c01, \text{true}, \text{true}, k) \Rightarrow \text{success} \quad [k = 0 \vee k = 1] \quad \}$$

BV-LCTRS Version

LCTRS

$$\Sigma = \Sigma_{theory}^{int} \cup \left\{ \begin{array}{l} \text{state} : Loc \times Loc \times bool \times bool \times bitvec_1 \Rightarrow State \\ c00 : Loc \\ c10 : Loc \\ c20 : Loc \\ crit0 : Loc \\ c01 : Loc \\ c11 : Loc \\ c21 : Loc \\ crit1 : Loc \\ \text{success} : State \\ \text{error} : State \end{array} \right\}$$

$$\left\{ \begin{array}{ll} 1 : & \text{state}(c00, p1, b0, b1, k) \rightarrow \text{state}(c10, p1, b01, b1, k) \quad [\neg(b01)] \\ 2 : & \text{state}(c10, p1, b0, b1, k) \rightarrow \text{state}(c20, p1, b0, b1, k) \quad [k \neq \#b0] \\ 3 : & \text{state}(c10, p1, b0, b1, k) \rightarrow \text{state}(crit0, p1, b0, b1, k) \quad [k = \#b0] \\ 4 : & \text{state}(c20, p1, b0, b1, k) \rightarrow \text{state}(c10, p1, b0, b1, k1) \quad [b1 \wedge k1 = \#b0] \\ 5 : & \text{state}(crit0, p1, b0, b1, k) \rightarrow \text{state}(c00, p1, b01, b1, k) \quad [b01] \\ 6 : & \text{state}(p0, c01, b0, b1, k) \rightarrow \text{state}(p0, c11, b0, b11, k) \quad [\neg(b11)] \\ 7 : & \text{state}(p0, c11, b0, b1, k) \rightarrow \text{state}(p0, c21, b0, b1, k) \quad [k \neq \#b1] \\ 8 : & \text{state}(p0, c11, b0, b1, k) \rightarrow \text{state}(p0, crit1, b0, b1, k) \quad [k = \#b1] \\ 9 : & \text{state}(p0, c21, b0, b1, k) \rightarrow \text{state}(p0, c11, b0, b1, k1) \quad [b0 \wedge k1 = \#b1] \\ 10 : & \text{state}(p0, crit1, b0, b1, k) \rightarrow \text{state}(p0, c01, b0, b11, k) \quad [b11] \\ 11 : & \text{state}(p0, p1, b0, b1, k) \rightarrow \text{success} \\ 12 : & \text{state}(crit0, crit1, b0, b1, k) \rightarrow \text{error} \end{array} \right\}$$

APR Problem

$$\{ \quad 13 : \quad \text{state}(c00, c01, \text{true}, \text{true}, k) \Rightarrow \text{success} \quad [k = \#b0 \vee k = \#b1] \quad \}$$